

Elastomeric Bearing Machining Guidelines

Specialized machining guidelines for Thordon, Feroform, and composite water-lubricated bearings — critical components for vertical turbine pumps and marine tail-shafts.

Specifications

Outside Diameter	70 mm to 700 mm
Length	Up to 500 mm
Clearance Tolerance	Down to 10 microns
Surface Finish	Optimized for bearing longevity

Machining Considerations

- ❗ Elastomeric materials flex and generate heat during machining
- ❗ Extremely sharp tooling with specific rake angles required
- ❗ Controlled coolant application to prevent material distortion
- ❗ Precise feed rates to maintain tight running clearances

Water Lubrication Grooves

We mill or broach straight or spiral water lubrication grooves into the bearing ID, ensuring proper fluid distribution for water-lubricated applications.

Assembly Service

- ❗ Interference fit calculation for bearing-to-ring press fitting
- ❗ Hydraulic press assembly of bearing into retainer ring
- ❗ Final bore check post-press to verify running clearance

Materials

- ❗ Thordon — primary composite for water-lubricated pumps
- ❗ Feroform F363 — high-performance composite
- ❗ Gunmetal / Bronze — traditional bearing material for marine
- ❗ Super Duplex Steel — corrosion-resistant retainer rings

Contact

Email sales@pakoshaft.com for bearing machining and assembly quotations.

